RAJALAKSHMI ENGINEERING COLLEGE

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M2

Started on	Monday, 28 September 2026, 10:07 PM
State	Finished
Completed on	Monday, 28 September 2026, 10:07 PM
Time taken	51 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <string.h>

int main() {
    char a[100], b[100];
    scanf("%s %s", a, b);

    int m = strlen(a), n = strlen(b);
    int dp[m + 1][n + 1];

    for (int i = 0; i <= m; i++)
        for (int j = 0; j <= n; j++)
            if (i == 0 || j == 0)
                dp[i][j] = 0;
            else if (a[i-1] == b[j-1])
                dp[i][j] = dp[i-1][j-1] + 1;
            else
                dp[i][j] = dp[i-1][j] > dp[i][j-1] ?
                           dp[i-1][j] : dp[i][j-1];

    printf("%d", dp[m][n]);
    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔

	Input	Expected	Got	
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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